

KNYSNA, South Africa

WMO: 689350

Lat: 34.0490S

Lon: 23.0814E

Elev: 54

StdP: 100.68

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.0	7.0	1.4	4.2	15.8	2.9	4.7	13.1	8.9	13.1	7.6	14.1	0.2	170	0.297

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	7.2	30.4	20.1	28.4	20.0	26.8	19.8	22.6	27.2	21.9	26.0	21.2	25.0	2.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.1	15.9	24.8	20.5	15.2	24.0	19.9	14.7	23.4	66.8	27.2	64.2	26.2	61.9	25.0	26.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.1	5.8	4.9	4.6	36.3	0.8	1.7	4.0	37.6	3.6	38.6	3.2	39.5	2.6	40.8		
(5)			3.7	24.1	0.7	1.4	3.2	25.1	2.8	25.9	2.4	26.6	1.8	27.6		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.8	21.7	22.1	20.7	18.4	16.5	14.4	14.0	14.2	15.4	17.3	18.6	20.6		
	DBStd	3.68	2.03	2.11	2.33	2.62	2.43	2.47	2.63	2.33	2.24	2.44	2.13	2.10		
	HDD10.0	1	0	0	0	0	0	0	1	1	0	0	0	0		
	HDD18.3	662	1	1	5	29	68	121	138	132	93	49	21	4		
	CDD10.0	2849	363	338	331	253	201	133	125	129	161	226	259	329		
	CDD18.3	468	106	105	77	32	11	4	4	3	4	17	29	76		
	CDH23.3	1829	394	392	268	139	78	37	42	22	31	84	82	260		
	CDH26.7	409	76	90	62	36	20	9	8	5	11	27	14	49		
(14) Wind	WSAvg	1.6	1.7	1.6	1.5	1.3	1.3	1.4	1.5	1.6	1.7	1.9	1.8	1.8		
(15) Precipitation	PrecAvg	725	50	45	61	66	62	62	54	75	62	75	60	54		
	PrecMax	1154	149	125	212	223	188	189	152	221	241	208	159	168		
	PrecMin	533	16	9	4	13	5	7	6	10	6	10	8	0		
	PrecStd	140	28	26	38	45	41	45	36	51	56	50	36	36		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	31.4	32.9	31.9	30.5	29.2	28.2	27.9	26.1	27.4	31.0	28.6	30.8		
		MCWB	21.7	22.9	21.2	19.4	16.7	15.6	15.2	14.8	16.4	18.9	19.2	20.2		
	2%	DB	28.6	29.1	28.2	26.9	25.3	23.7	23.8	22.0	23.0	25.5	25.5	27.5		
		MCWB	21.4	21.5	20.6	18.9	16.8	14.5	14.1	14.7	16.0	17.7	19.2	20.6		
	5%	DB	26.7	26.9	26.1	24.4	22.4	21.0	20.6	19.8	20.8	22.8	23.6	25.8		
		MCWB	20.8	20.8	20.2	18.2	16.4	14.4	13.7	14.0	15.2	17.2	18.6	19.9		
	10%	DB	25.4	25.5	24.4	22.5	20.5	19.0	18.5	18.2	19.1	21.1	22.2	24.5		
		MCWB	20.2	20.4	19.6	17.8	15.8	14.0	13.4	13.5	14.6	16.7	17.9	19.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.3	23.8	22.7	21.2	19.7	17.3	16.9	17.2	18.2	20.4	21.5	22.9		
		MCDB	28.3	29.8	27.7	26.8	25.0	23.0	22.6	22.3	23.5	26.6	25.2	27.2		
	2%	WB	22.3	22.6	21.6	20.1	18.1	16.1	15.6	15.9	16.9	18.8	20.3	21.8		
		MCDB	26.8	27.2	26.1	24.8	22.5	20.8	20.4	20.0	20.7	23.6	23.7	25.6		
	5%	WB	21.6	21.8	20.9	19.2	17.0	15.2	14.6	14.9	16.0	17.9	19.3	20.9		
		MCDB	25.6	25.5	25.0	23.1	21.0	19.8	19.1	18.7	19.8	21.8	22.6	24.4		
	10%	WB	20.9	21.1	20.2	18.3	16.2	14.3	13.8	14.1	15.1	17.1	18.5	20.1		
		MCDB	24.6	24.4	23.7	21.7	19.7	18.5	17.9	17.6	18.6	20.4	21.7	23.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.1	7.2	7.4	8.0	8.8	9.9	9.4	8.7	7.7	7.2	6.9	7.2		
		MCDBR	9.7	9.9	10.6	11.8	12.3	13.5	13.1	11.9	11.0	10.5	9.6	9.6		
		MCWBR	4.3	4.2	4.5	5.3	6.1	6.9	6.6	6.0	5.7	5.0	4.6	4.4		
	5% WB	MCDBR	8.2	8.3	8.7	9.9	10.2	11.3	10.9	10.4	9.9	9.2	8.1	7.9		
		MCWBR	4.0	3.9	4.2	5.0	5.9	6.7	6.4	6.0	5.7	4.9	4.3	4.1		
(40) Clear-Sky Solar Irradiance	taub		0.381	0.385	0.371	0.344	0.329	0.317	0.319	0.350	0.366	0.387	0.371	0.378		
	taud		2.477	2.471	2.497	2.556	2.574	2.582	2.570	2.473	2.410	2.386	2.468	2.472		
	Ebn at Noon		956	931	907	878	838	824	840	856	896	918	960	963		
	Edh at Noon		117	113	103	87	75	70	74	92	110	122	117	118		
(44) All-Sky Solar Radiation	RadAvg		7.45	6.66	5.45	4.08	3.06	2.62	2.87	3.70	5.00	6.08	7.16	7.72		
	RadStd		0.28	0.23	0.28	0.17	0.16	0.14	0.18	0.17	0.23	0.31	0.35	0.43		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)		N/A	N/A	+1.88	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page